# MSPM0 USB-C controller board

A general-purpose USB-C powered MSPM0G3507 board I designed in KiCad, with a USB-to-UART bridge for talking to a PC and SWD for programming.

Rev A (schematic) pcb embedded usb-c kicad

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|---|---|---|---|
| Author | Sao Aphisith Sithisack | Revision | Rev A (schematic) |
| Date | October 12, 2026 | Tools | KiCad |
| Microcontroller | MSPM0G3507 | Supply | 3.3 V from USB-C |
| Status | schematic captured | Online | https://saosith.github.io/projects/mspm0-usb-c-controller-board/ |

**Design only.** Status: schematic complete in KiCad. Layout, fabrication and bring-up come next, so no values on this page are measured.
VBUS = 4.75 to 5.25 V (USB 2.0 range), room temperature

Abstract

This is a general-purpose microcontroller board for my projects. It runs from a laptop's USB-C port, appears to the PC as a serial port, and can be reprogrammed on the bench without extra hardware. The design centres on a TI MSPM0G3507 microcontroller with a CH340E USB-to-UART bridge, USB ESD protection, a 3.3 V LDO, an 8 MHz crystal and a Tag-Connect SWD footprint. Status: schematic complete in KiCad. Next: layout, fabrication and bring-up.

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Electrical characteristics

| Parameter | Sym. | Target | Simulated | Unit | vs. target |
|---|---|---|---|---|---|
| 3.3 V rail | V_3V3 | 3.3 ± 3% | – | V | — |
| Total 3.3 V load current [1] | I_load | ≤ 100 | – | mA | — |
| LDO dissipation at full load [2] | P_LDO | ≤ 0.2 | – | W | — |
| UART baud-rate error (115200) | | < 1 | – | % | — |
| ADC reference noise [3] | | < 1 LSB (0.8 mV) | – | | — |
| USB ESD, D+ and D– [4] | | ±8 kV contact | – | | — |

Conditions VBUS = 4.75 to 5.25 V (USB 2.0 range), room temperature

1 One USB 2.0 unit load, so the board works on any port before enumeration.

2 P = (5.25 V − 3.3 V) × 0.1 A ≈ 0.2 W, the worst case for the SPX3819 LDO.

3 12-bit ADC with a 3.3 V reference: 3.3 V / 4096 ≈ 0.8 mV per step.

4 IEC 61000-4-2 contact discharge, the level the USBLC6-2 array is rated for.

## Introduction

I wanted a general-purpose microcontroller board I could reuse across projects. I wanted it to run off my laptop's USB-C port, send data to the PC over a serial port, and be reprogrammed on the bench without extra hardware. So I designed this board around a TI MSPM0G3507 (Arm Cortex-M0+) [1]. It has USB-C power, a USB-to-UART bridge, a crystal, an SWD debug pad and a filtered ADC reference.

**Status:** schematic complete. **Next:** layout, then fabrication and bring-up. No values on this page are measured.

## Requirements

- **Power** from a laptop's USB-C port, with no separate bench supply.
- **Serial link** to a PC for logging, with no extra adapter.
- **Reprogramming** on the bench without extra hardware on the board.

- **Clean ADC reference**, so analog measurements are not limited by reference noise.

The table at the top lists the numeric targets. They come from the USB 2.0 limits and the part datasheets; nothing is measured yet.

## System overview

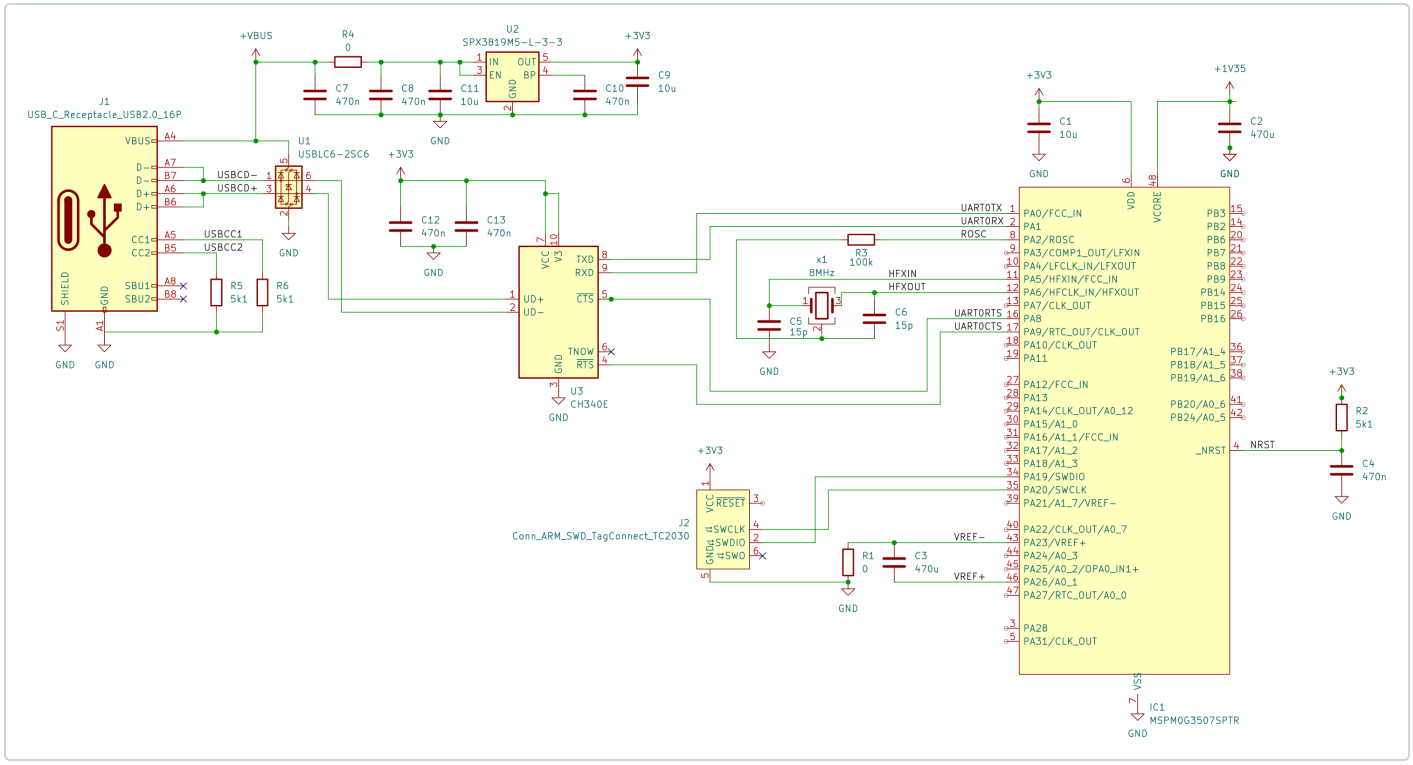



**Fig. 1** The full schematic. Power comes in on the left, the microcontroller is on the right, and the USB-to-UART bridge is in the middle.

## Design

### Power path

USB-C supplies 5 V on VBUS. Normally, a USB-C host or charger keeps VBUS off until it detects a device. **R5 and R6 pull CC1 and CC2 to ground through 5.1 kΩ**, the value the USB Type-C specification sets for a sink [5], so the host sees the board as a device requesting power. VBUS then passes through R4, a 0 Ω link that can be replaced with a fuse

or removed to cut the supply. An SPX3819 low-dropout regulator (LDO) [4] drops the 5 V to 3.3 V and dissipates the difference as heat.

### Data path and protection

D+ and D− pass through a USBLC6-2SC6 ESD array [3] first. If someone touches the connector, the electrostatic discharge (ESD) is clamped to ground before it reaches the CH340E USB-to-UART bridge [2]. In layout, I'll place the array right at the connector so the discharge has the shortest path to ground. The CH340E appears to the PC as a standard serial port and to the microcontroller as a two-wire UART.

### Clock, reset, reference

An 8 MHz crystal with two 15 pF load capacitors drives HFXIN/HFXOUT. A crystal is typically accurate to tens of ppm, while an on-chip RC oscillator is usually off by around a percent, which is enough to disturb UART timing. The pull-up and capacitor on NRST hold the microcontroller in reset until the 3.3 V rail is stable. A 100 kΩ resistor on ROSC sets the reference current for the internal oscillator.

An RC low-pass filter sits on the ADC reference. The ADC measures every input as a fraction of this reference voltage, so noise on the reference appears in every reading.

Programming and debugging use a TC2030 Tag-Connect footprint: the cable's spring pins press onto pads on the board, so no header is soldered on.

## Design decisions

| Decision | Options considered | Choice | Why | Status |
|---|---|---|---|---|
| Clock source | Internal oscillator, or an external 8 MHz crystal | **8 MHz crystal (two 15 pF load caps)** | A crystal is typically accurate to tens of ppm, while an on-chip RC oscillator is usually off by around a percent. I chose the crystal so the UART baud rate and any timing-critical measurements do not depend on that error. | decided |
| Programming connector | Soldered pin header, or a TC2030 Tag-Connect footprint | **TC2030 Tag-Connect** | The Tag-Connect cable's spring pins press onto bare pads, so the board needs no header. I chose it to save board area and one part. | decided |
| Crystal load capacitors | 15 pF as drawn, or a value recomputed from the crystal's rated load capacitance | **Not settled** | To verify before layout: C = 2 × (CL − Cstray), using the chosen crystal's rated load capacitance CL and an estimate of the board's stray capacitance. | open |

## Project status

Sections for layout, simulation and test results are added as each stage is completed.

| Stage | Status | Plan |
|---|---|---|
| Schematic | ✓ Complete | — |
| Design review | ▶ In progress | Check decoupling, the serial handshake wiring and the ADC reference filter against the MSPM0 [1] and CH340E [2] datasheets. |
| Simulation | ○ Planned | ADC reference filter response and LDO start-up. |
| Layout | ○ Planned | Stack-up, placement and routing, with the ESD array right at the USB-C connector. |
| Fabrication | ○ Planned | — |
| Bring-up and test | ○ Planned | Rail voltages, USB enumeration, UART loop-back and SWD flashing. |

## Open issues

- **No auto-reset.** The bridge's handshake lines are not connected to NRST, so firmware loads through SWD and the PC cannot reset the microcontroller. Wiring RTS/DTR to NRST is an option for Rev B.
- **Design review before layout.** Check decoupling, the serial handshake wiring and the ADC reference filter against the MSPM0 [1] and CH340E [2] datasheets.
- **Crystal load capacitors.** Recalculate with $C = 2 \times (C_L - C_{stray})$ once the crystal part number is fixed.

## Revision history

| Rev | Date | What changed |
|---|---|---|
| A | `2026-09-10` | Schematic captured in KiCad and first written up. |
| A | `2026-10-12` | Write-up restructured as a report: abstract, decisions, BOM appendix, placeholders for layout and test. |

## References


1. Texas Instruments, MSPM0G350x mixed-signal microcontroller datasheet
2. WCH, CH340 USB-to-serial bridge datasheet
3. STMicroelectronics, USBLC6-2 ESD protection datasheet
4. MaxLinear, SPX3819 500 mA low-dropout regulator datasheet
5. USB Type-C Cable and Connector Specification (sink Rd pull-down on CC1/CC2)

# Appendix: Bill of materials

| Ref | Part | Value | Package | Source |
|---|---|---|---|---|
| IC1 | MSPM0G3507SPTR microcontroller | — | LQFP-48 7x7 mm | LCSC C22362630 |
| U2 | SPX3819M5-L-3-3 LDO regulator | 3.3 V | SOT-23-5 | TBD |
| U3 | CH340E USB-to-UART bridge | — | MSOP-10 | TBD |
| U4 | LIS2DH accelerometer | — | LGA-14 2x2 mm | TBD |
| D1 | USBLC6-2SC6 ESD array | — | SOT-23-6 | TBD |
| J1 | USB-C receptacle (USB 2.0, 16 pin) | — | GCT USB4105 footprint | TBD |
| x1 | Crystal | 8 MHz | 3225 4-pin | TBD |
| C1, C2, C4, C6, C7, C13 | Capacitor | 470 nF | 0201 | TBD |
| C3, C5, C11 | Capacitor | 10 µF | 0805 | TBD |
| C8, C9 | Capacitor | 15 pF | 0201 | TBD |
| C10, C12, C14 | Capacitor | 470 µF (as drawn) | 0201 | TBD |
| R1, R2, R6–R8 | Resistor | 5.1 kΩ | 0201 | TBD |
| R3, R4 | Resistor | 0 Ω | 0201 | TBD |
| R5 | Resistor | 100 kΩ | 0201 | TBD |